MuntsOS Embedded Linux

Application Note #20: Orange Pi Zero 2W Target Platform Notes

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Introduction

The Orange Pi Zero 2W is a small Linux microcomputer with a form factor very similiar to the Raspberry Pi Zero 2 W, making it ideal for embedded system projects. It has a 1500 MHz Allwinner H618 ARMv8 Cortex-A53 quad-core CPU and comes with 1 to 4 GB of RAM and on-board Bluetooth and WiFi radios. It also as a 40-pin expansion header highly compatible with that of Raspberry Pi boards.

Standard Hardware Configuration

After installing a **MuntsOS Embedded Linux** Thin Server, the Orange Pi Zero 2W will have **`I2C1`**, **`I2C2`**, **`PWM0`**, **`SPI0`**, and **`UART0`** hardware subsystems active. Unlike the Raspberry Pi, which can map **`PWM0`** to either **`GPIO12`** or **`GPIO18`**, the Orange Pi Zero 2W can only map **`PWM0`** to **`GPIO12`**.

		3.3V	1	o o	2	5V		
I2C1 SDA	GPIO2	PI8	3	o o	4	5V		
I2C1 SCL	GPIO3	PI7	5	o o	6	GND		
PWM2	GPIO4	PI13	7	o o	8	PH0	GPIO14	TXD0
		GND	9	o o	10	PH1	GPIO15	RXD0
	GPIO17	PH2	11	o o	12	PI1	GPIO18	
	GPIO27	PH3	13	o o	14	GND		
	GPIO22	PI5	15	o o	16	PI14	GPIO23	PWM3
		3.3V	17	o o	18	PH4	GPIO24	
SPI0 MOSI	GPIO10	PH7	19	o o	20	GND		
SPI0 MISO	GPIO9	PH8	21	o o	22	PI6	GPIO25	
SPI0 SCLK	GPIO11	PH6	23	o o	24	PH5	GPIO8	SPI0 $\overline{CE0}$
		GND	25	o o	26	PH9	GPIO7	SPI0 $\overline{CE1}$
I2C2 SDA	GPIO0	PI10	27	o o	28	PI9	GPIO1	I2C2 SCL
	GPIO5	PI0	29	o o	30	GND		
	GPIO6	PI15	31	o o	32	PI11	GPIO12	PWM0
PWM1	GPIO13	PI12	33	o o	34	GND		
	GPIO19	PI2	35	o o	36	PC12	GPIO16	
	GPIO26	PI16	37	o o	38	PI4	GPIO20	
		GND	39	o o	40	PI3	GPIO21	
1	2	3				4	5	6

Column 1, 2, 5, and 6 pin names match Raspberry Pi GPIO header pin names.
Column 3 and 4 pin names match the Orange Pi Zero 2W schematic diagram.

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Alternate GPIO pin functions (shown as gray in the above diagram) can be configured with device tree overlays described in the following section.

Installing U-Boot

See https://docs.u-boot.org/en/v2026.04/board/allwinner/sunxi.html for more information.

The easiest way to boot **MuntsOS Embedded Linux** on an Orange Pi Zero 2W is to install the boot loader image file **`u-boot-sunxi-with-spl.bin`** to the on-board SPI flash using the **`sunxi-fel`** program (provided by the **`sunxi-tools`** package from the Debian distribution).

1. Download **`bootfiles.tgz`** from:

https://github.com/pmunts/muntsos/blob/master/boot/OrangePiZero2W/bootfiles.tgz

2. Unpack **`bootfiles.tgz`**:

```
tar xzf bootfiles.tgz
```

3. Connect the endmost USB-C connector on your target board to your development host computer and run:

```
sudo sunxi-fel spiflash-write 0 u-boot/u-boot-sunxi-with-spl.bin
```

Device Tree Overlays

MuntsOS includes the following device tree overlays for altering the Orange Pi Zero 2W hardware configuration. Device tree overlays are applied by the boot loader and selected by editing **`/boot/config.txt`**, changing **`OVERLAYS=`** to *e.g.* **`OVERLAYS=disable-pwm`**.

`disable-i2c1.dtbo`

Disables the **`I2C1`** hardware subsystem and enables **`GPIO2`** and **`GPIO3`**.

`disable-i2c2.dtbo`

Disables the **`I2C2`** hardware subsystem and enables **`GPIO0`** and **`GPIO1`**. These pins are reserved on a Raspberry Pi.

`disable-spi1.dtbo`

Disables the **`SPI1`** hardware subsystem (which actually instantiates as **`/dev/spidev0.0`** and **`/dev/spidev0.1`**) and enables **`GPIO7`**, **`GPIO8`**, **`GPIO9`**, **`GPIO10`**, and **`GPIO11`**.

`disable-pwm.dtbo`

Disables the **`PWM0`** hardware system and enables **`GPIO12`**.